



SYDE252 - W01C – Signal Manipulation

Dr. Matt Borland

Problem 1

Drill 1.1 from the textbook...

Problem 2

Even and Odd Signals

A CT signal is even if and only if $x(t) = x(-t)$

A CT signal is odd if and only if $x(t) = -x(-t)$

A CT signal can be neither even nor odd, but it always has its even part and its odd part, defined as:

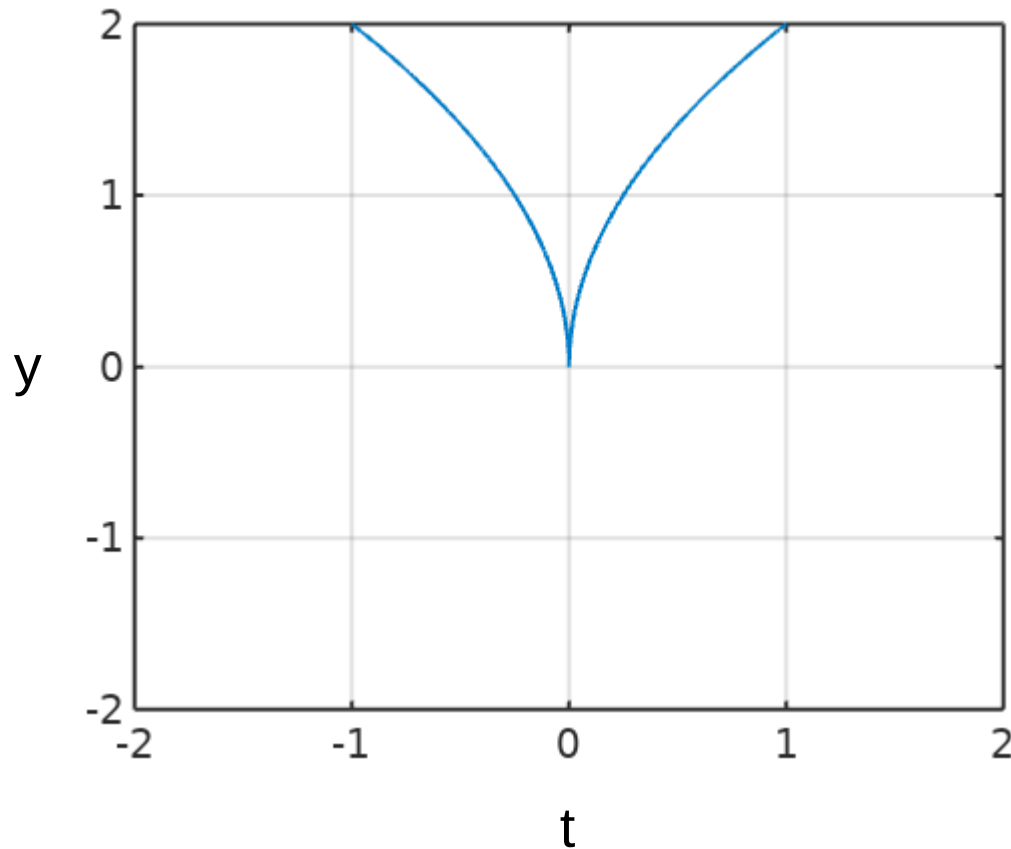
$$\mathcal{E}\{x(t)\} = \frac{1}{2}(x(t) + x(-t)) \quad \mathcal{O}\{x(t)\} = \frac{1}{2}(x(t) - x(-t))$$

And it is trivial to verify that:

$$x(t) = \mathcal{E}\{x(t)\} + \mathcal{O}\{x(t)\}$$

Problem 2

Even Signals

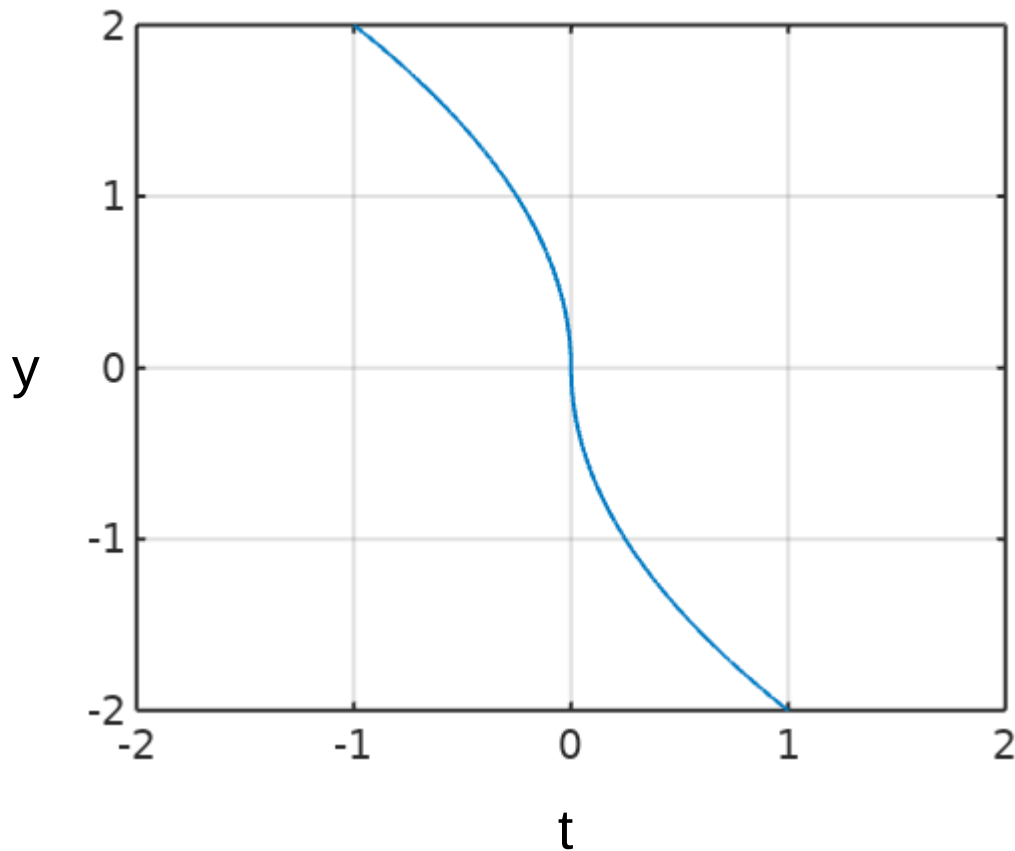


$$y(t) = y(-t)$$

```
6 # Even signal
7 t = -1:0.001:1;
8 x = zeros(size(t));
9 x(t>0) = 2*sqrt(t(t>0));
10 x(t<=0) = 2*sqrt(-t(t<=0));
```

Problem 2

Odd Signals

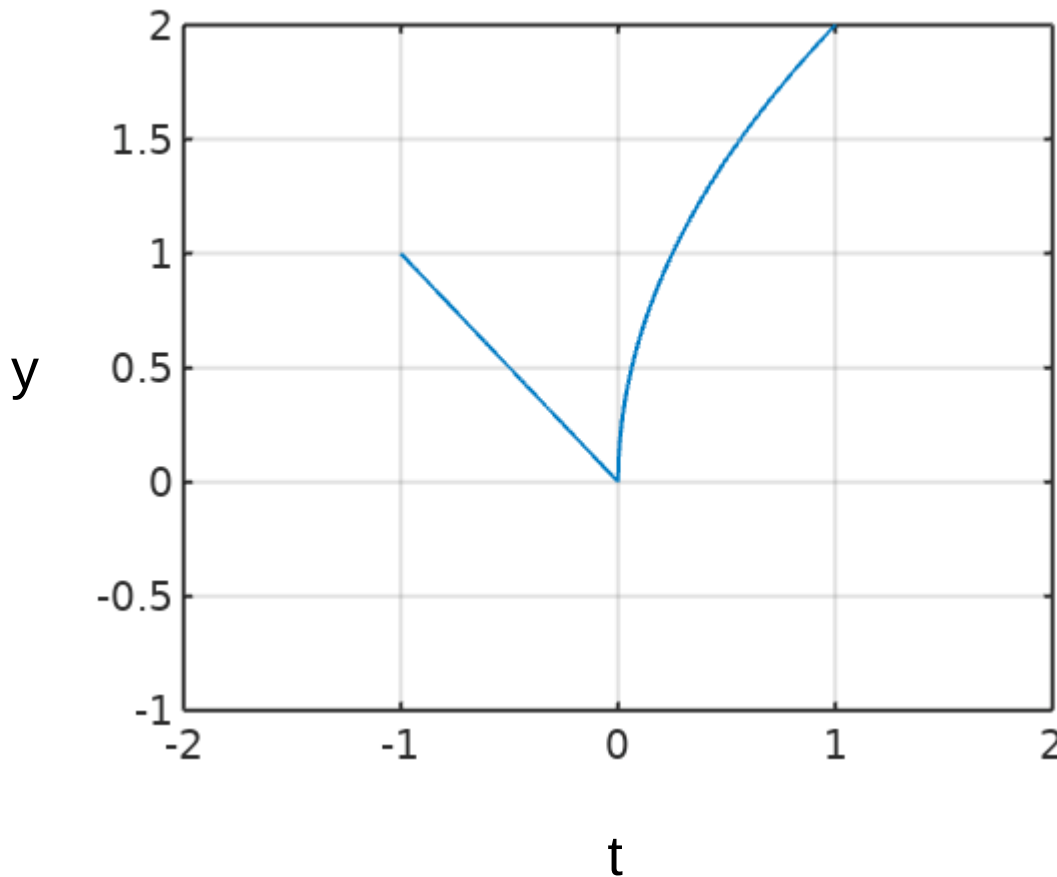


$$y(-t) = -y(t)$$

```
21 # Odd signal
22 t = -1:0.001:1;
23 x = zeros(size(t));
24 x(t>0) = -2*sqrt(t(t>0));
25 x(t<=0) = 2*sqrt(-t(t<=0));
```

Problem 2

Decomposition into Even and Odd Parts

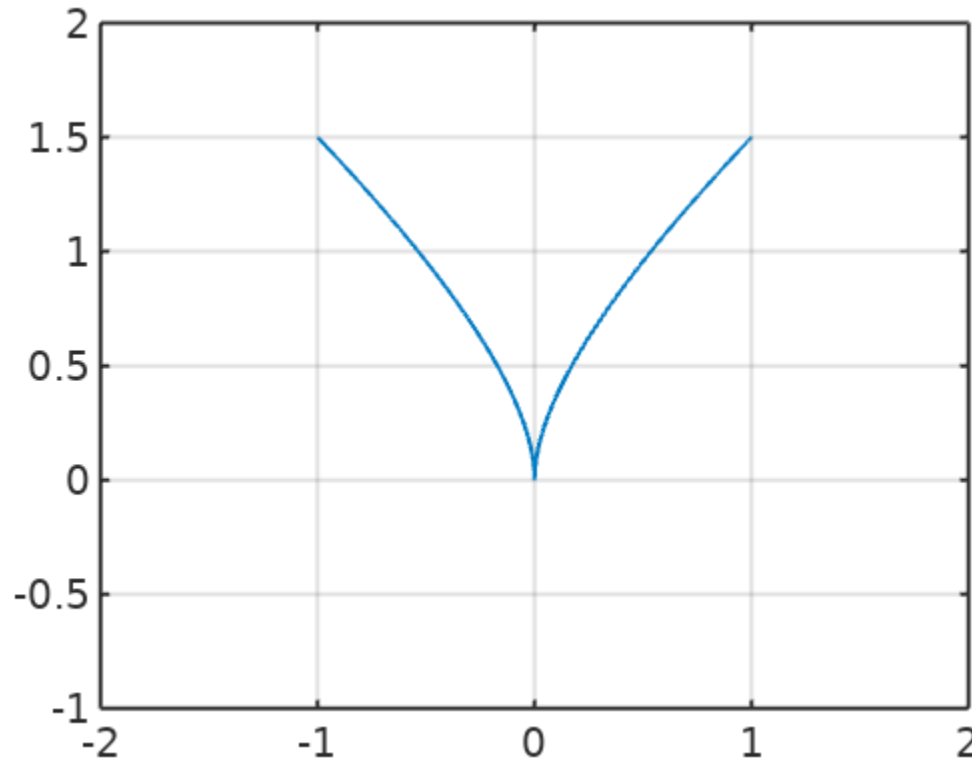


$$y(t) = \begin{cases} 2\sqrt{t} & \text{if } t > 0 \\ -t & \text{if } t \leq 0 \end{cases}$$

Problem 2

Decomposition into Even and Odd Parts

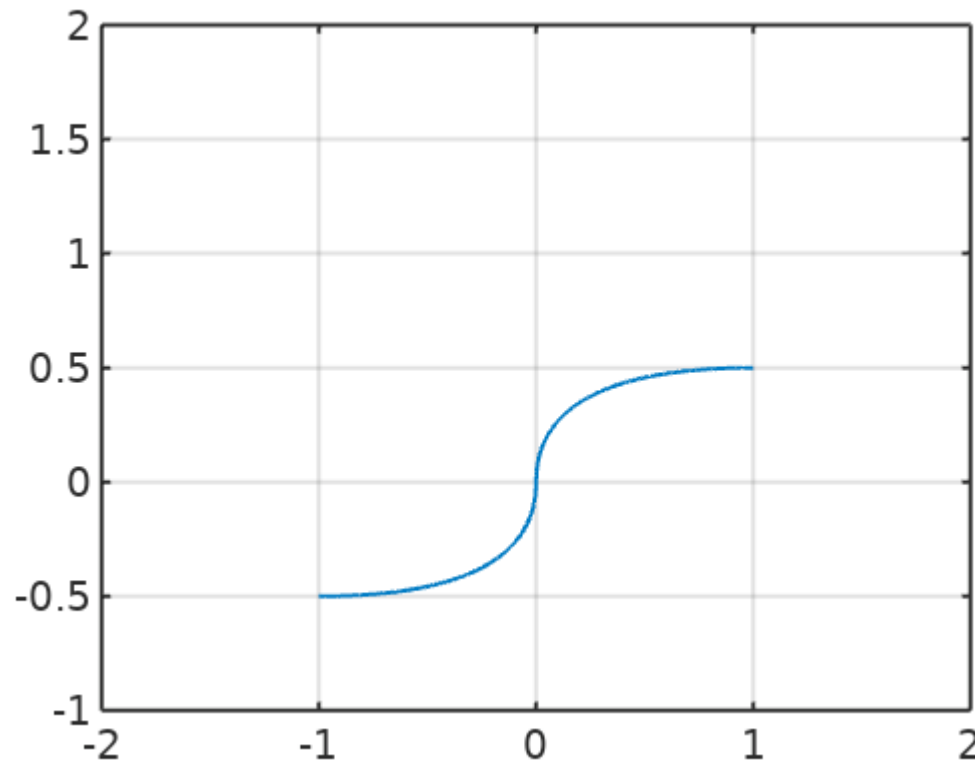
$$\text{Even}\{y(t)\} = \frac{1}{2}[y(t) + y(-t)]$$



Problem 2

Decomposition into Even and Odd Parts

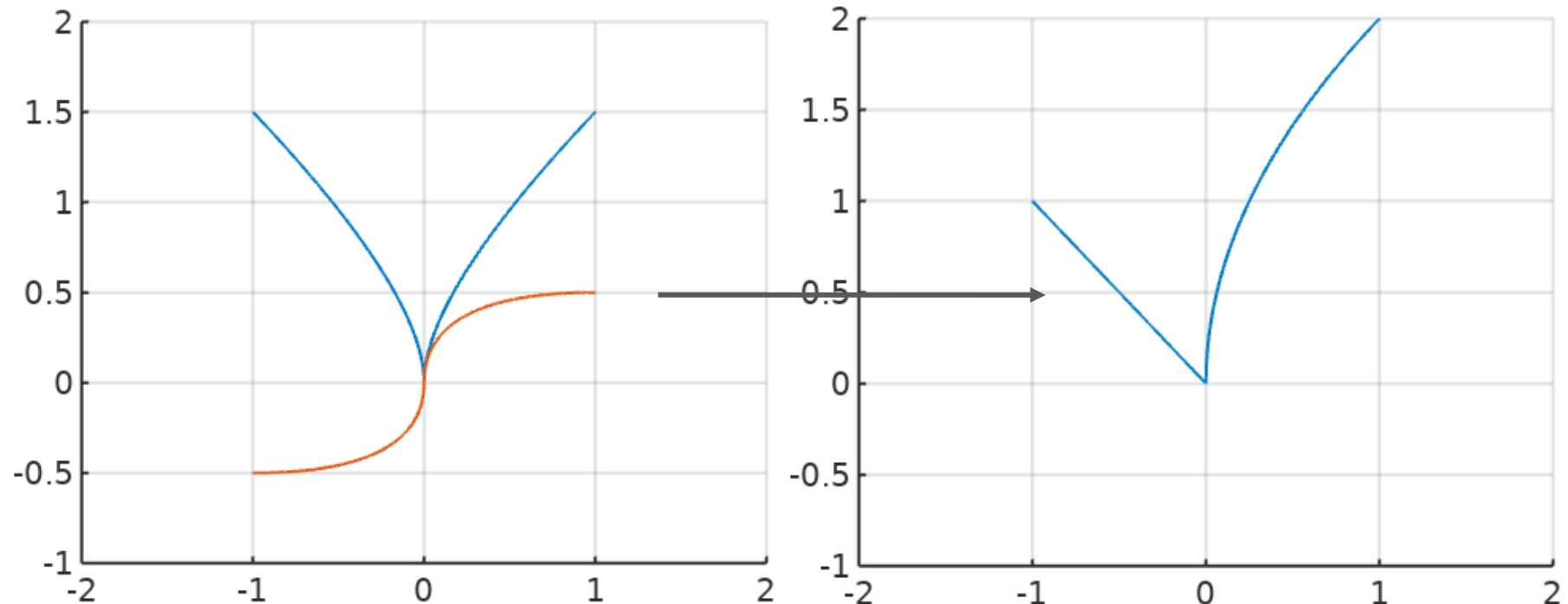
$$\text{Odd}\{y(t)\} = \frac{1}{2}[y(t) - y(-t)]$$



Problem 2

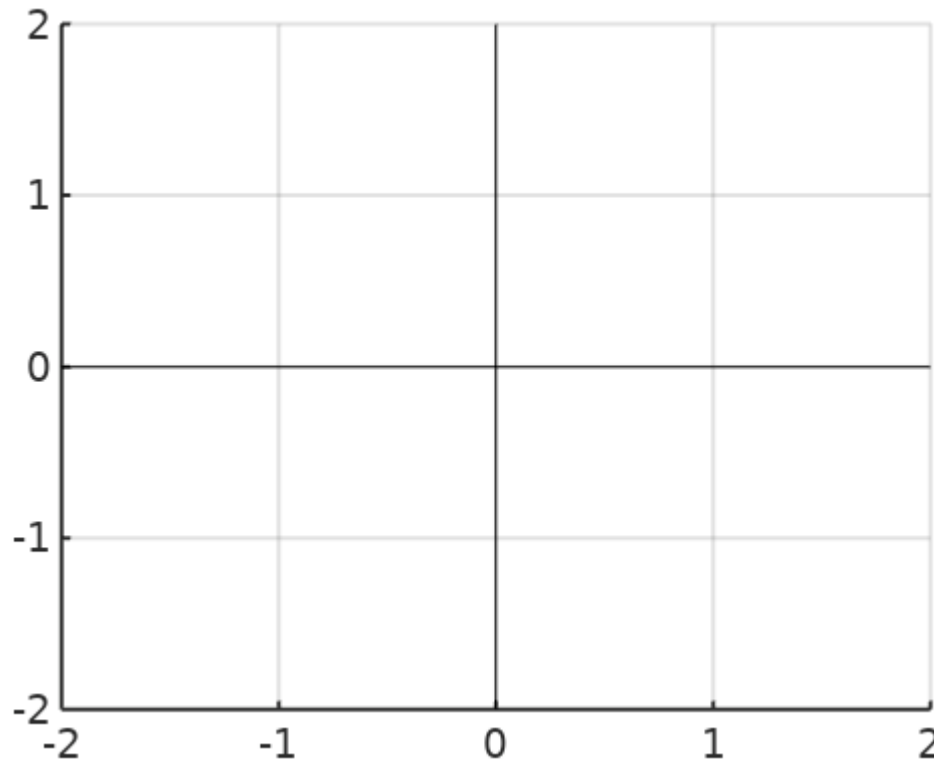
Decomposition into Even and Odd Parts

$$y(t) = \text{Even}\{y(t)\} + \text{Odd}\{y(t)\}$$



Problem 2

Draw 2 Even and 2 Odd Signals on this kind of grid:



Problem 3

A CT signal, $x(t)$, is **periodic** if and only if there exist a real number T , such that $x(t + T) = x(t), \forall t$. And T is call **a period** of $x(t)$, which is said to be periodic with period of T

If T is a period of a CT signal $x(t)$, any multiple of T , *i.e.* kT , where k is an integer, is also a period of $x(t)$.

Therefore, any periodic CT signal, $x(t)$, has infinite number of periods. And the smallest period, T_0 , is call **the fundamental period** of $x(t)$.

Problem 3

Determine if the CT signal defined below is ***periodic***.

$$x(t) = 3 \cos(4t + \pi/3)$$

Is there a T such that $x(t) = x(t + T)$?

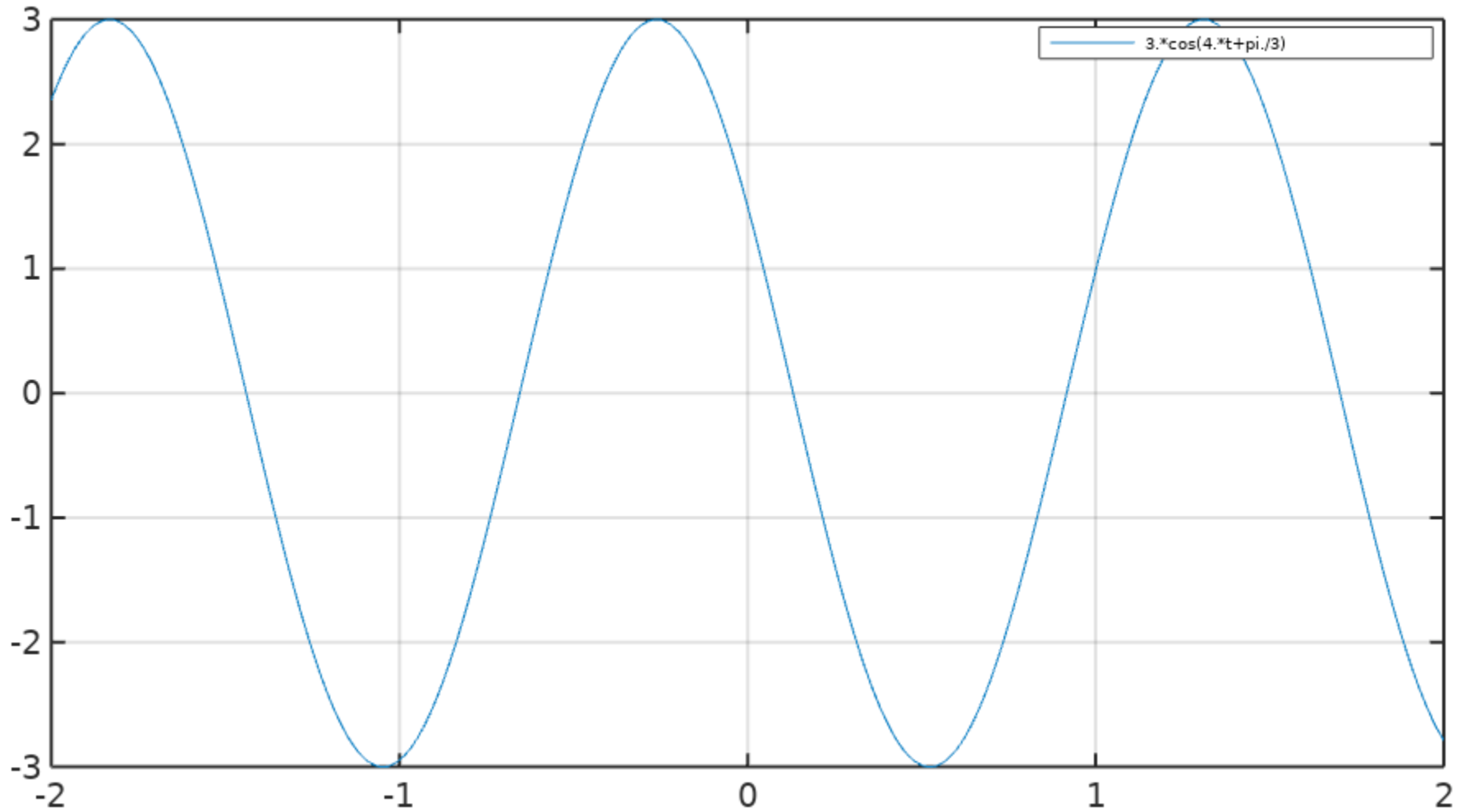
Problem 3

Octave Code to plot that function:

```
1 # Create a time vector
2 t = -5:0.01:5;
3
4 # Calculate the signal
5 x = 3*cos(4*t+pi/3);
6
7 # Create a figure window (#1) to plot the signal
8 figure(1);
9
10 # clear the figure in case something else has been drawn on it
11 clf;
12
13 # plot the independent variable, t, versus x,
14 # and use a green line for fun
15 plot(t,x,'g');
```

Try it out!

Problem 3



Problem 3

